

Post-retirement Age Mortality in Australia: A Quantification of Socio-Economic Differences

Andrés Villegas – School of Risk and Actuarial Studies, UNSW Sydney

Co-author: Francis Hui, Fei Huang

We thank the **Australian Government Actuary** for their invaluable support in completing this project.



Motivation

- The Australian superannuation system's retirement phase needs improvements to better supplement or replace the Age Pension.
- Few Australians invest in longevity products → limited mortality data
- Australian Life Table (ALT) lacks detail beyond age and gender
- Investigate mortality differences by socio-economic variable → Improve longevity pool design for enhanced retirement outcomes
- Partnership between Australian Government Actuary and UNSW.

Dataset

- Data sourced from Person Level Integrated Data Asset (PLIDA) via ABS DataLab.
- Collaborators: ABS, ATO, Department of Education, Health, Human Services, and Social Services.

Available Data

- **2016 Census Data:** Entire population snapshot as of August 9, 2016.
- **2009–2016 Social Security Related Information (SSRI):** Benefits, payments, marital status, education.
- **Demographics:** For individuals aged 50+ (post-Dec 2007). Includes gender, birth/death dates, indigenous status.

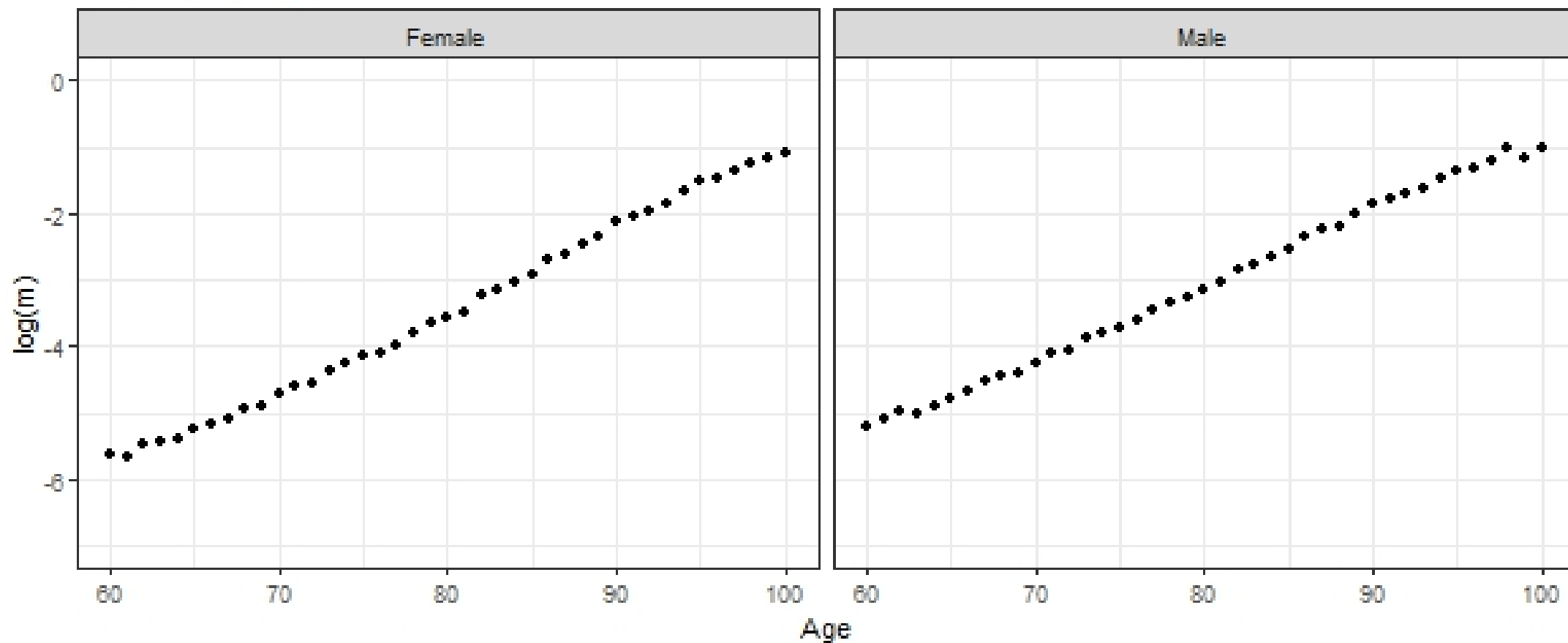
Our sample

- Follow 2016 census population for a year from 1 September 2016 and 31 August 2017
 - Maximise covariate information
 - Minimise death underreporting
 - Avoid issues with seasonality

Age range	Exposure		Deaths	
	Females	Males	Females	Males
60-64	621046.63	584730.65	2518	3823
65-69	561972.25	534151.86	3603	5614
70-74	443133.38	417478.67	4915	7445
75-79	328190.86	292705.65	6472	9076
80-84	241294.20	193058.22	9077	10766
85-89	171998.74	116084.92	12701	11902
90-94	87037.82	44106.01	12346	7924
95-100	24408.87	8790.14	6044	2506
Total	2479082.75	2191106.11	57676	59056

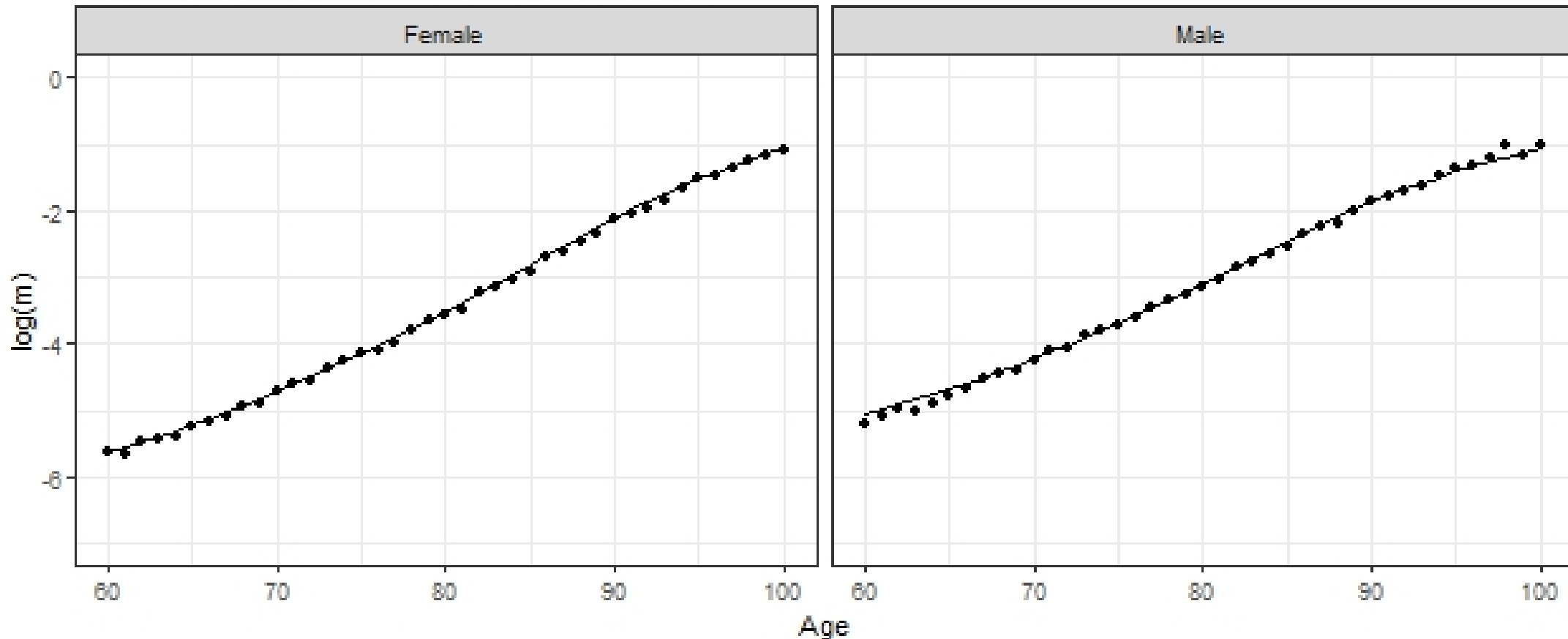
Our sample

Mortality by Age



Our sample matches ALT 2015-2017

Mortality by Age (ALT 2015-2017 solid line)



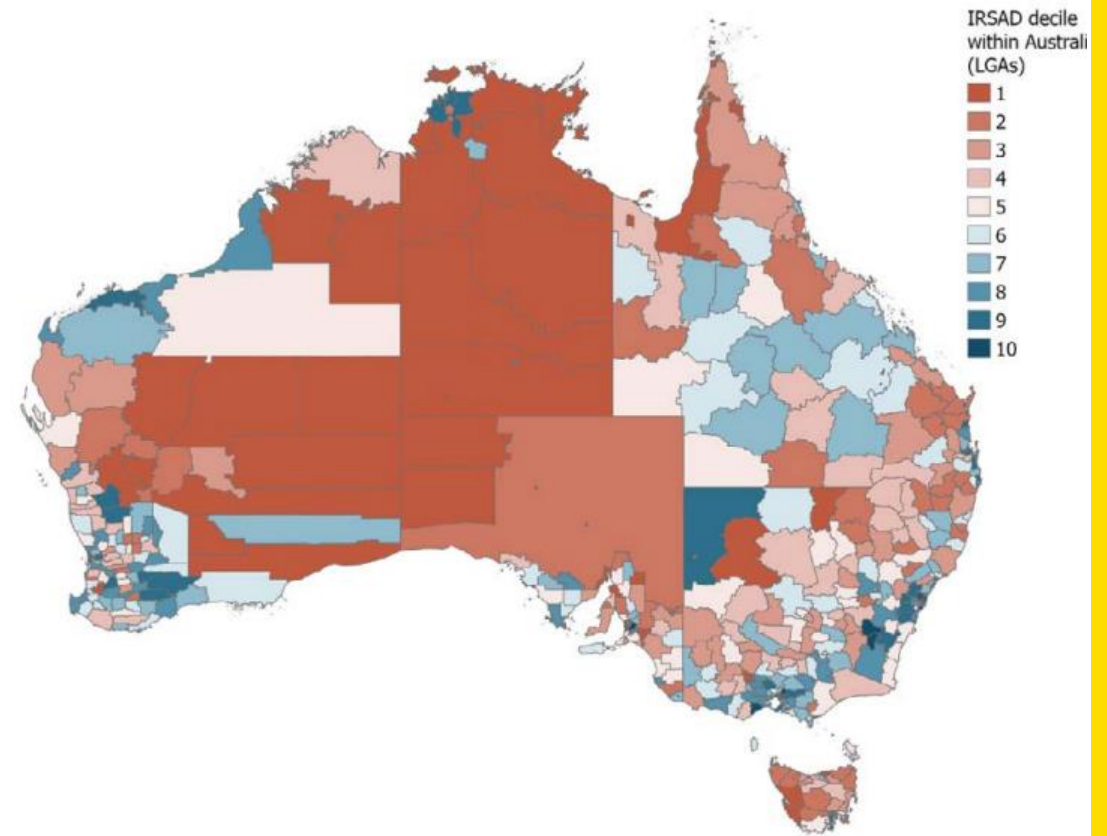
Covariates

- Index of Relative Socio-economic Advantage and Disadvantage (IRSAD) -- one of the SEIFA variables
- Marital Status
- Income
- Homeownership

Note: Education and Occupation are not considered as they are noisy and contains significant missing data for the retirement age population. SEIFA already contains information about education and employment.

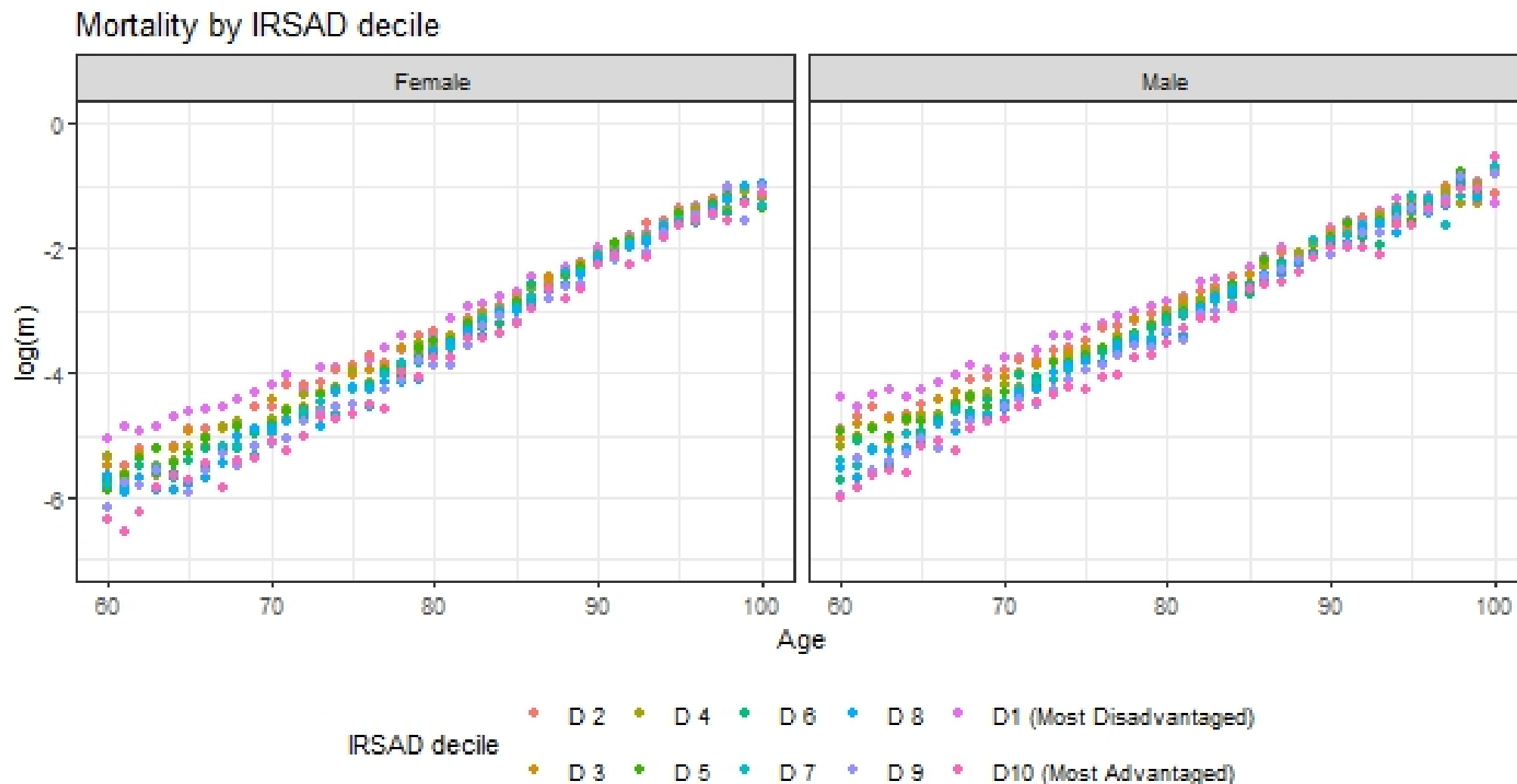
Index of Relative Socio-economic Advantage and Disadvantage (IRSAD)

- One of the SEIFA indices, summarises economic and social conditions in an area by measuring both advantage and disadvantage at SA1 level based on “SA1_IRSAD_2016” code of census.
- SA1_IRSAD_2016 provides 10 decile levels ranked from D1 (most disadvantaged) to D10 (most advantaged).



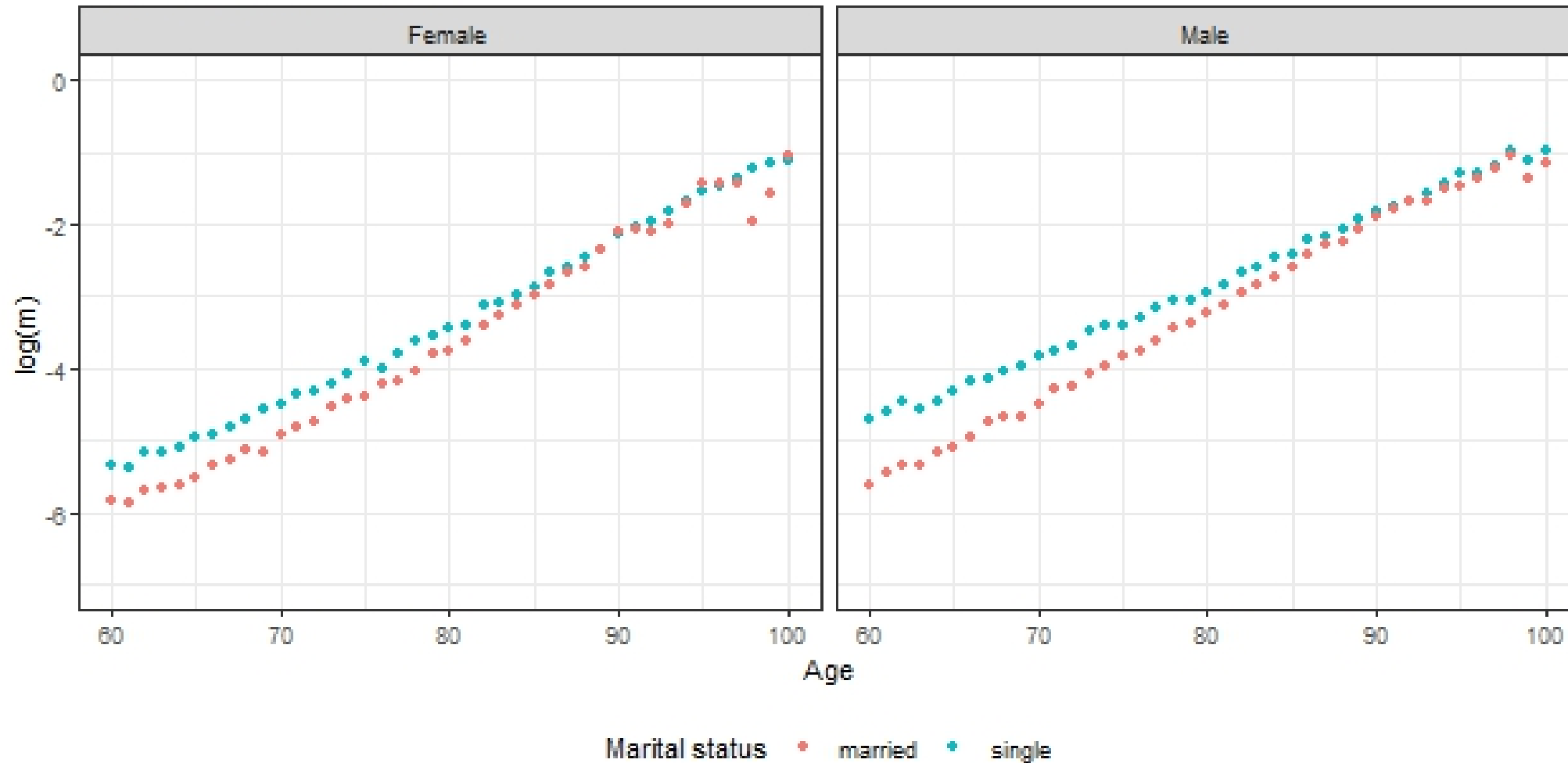
Source: BCARR analysis of the ABS Index of Relative Socio-Economic Advantage and Disadvantage (IRSAD), 2021.

Index of Relative Socio-economic Advantage and Disadvantage (IRSAD)



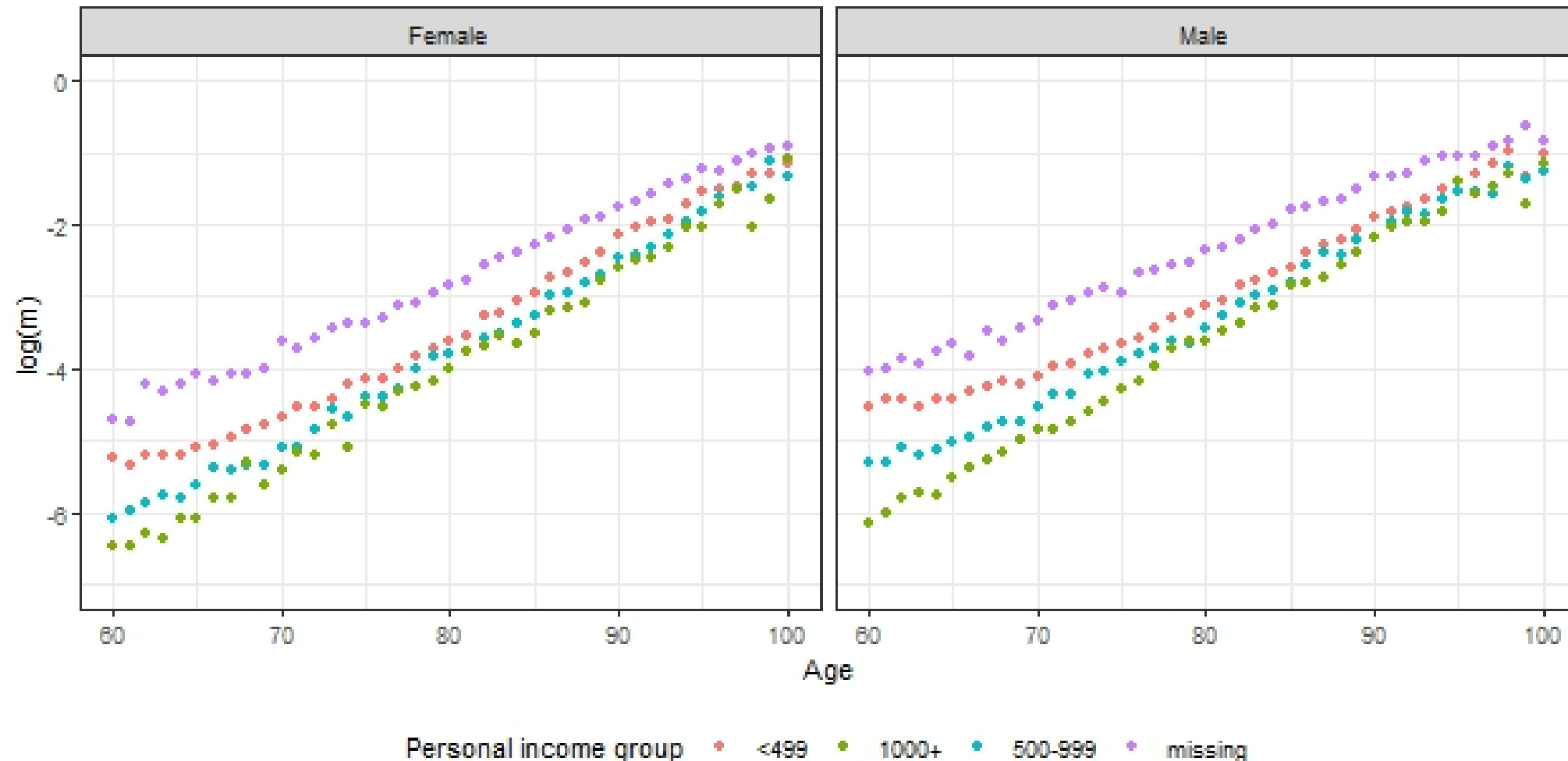
Marital Status

Mortality by marital status

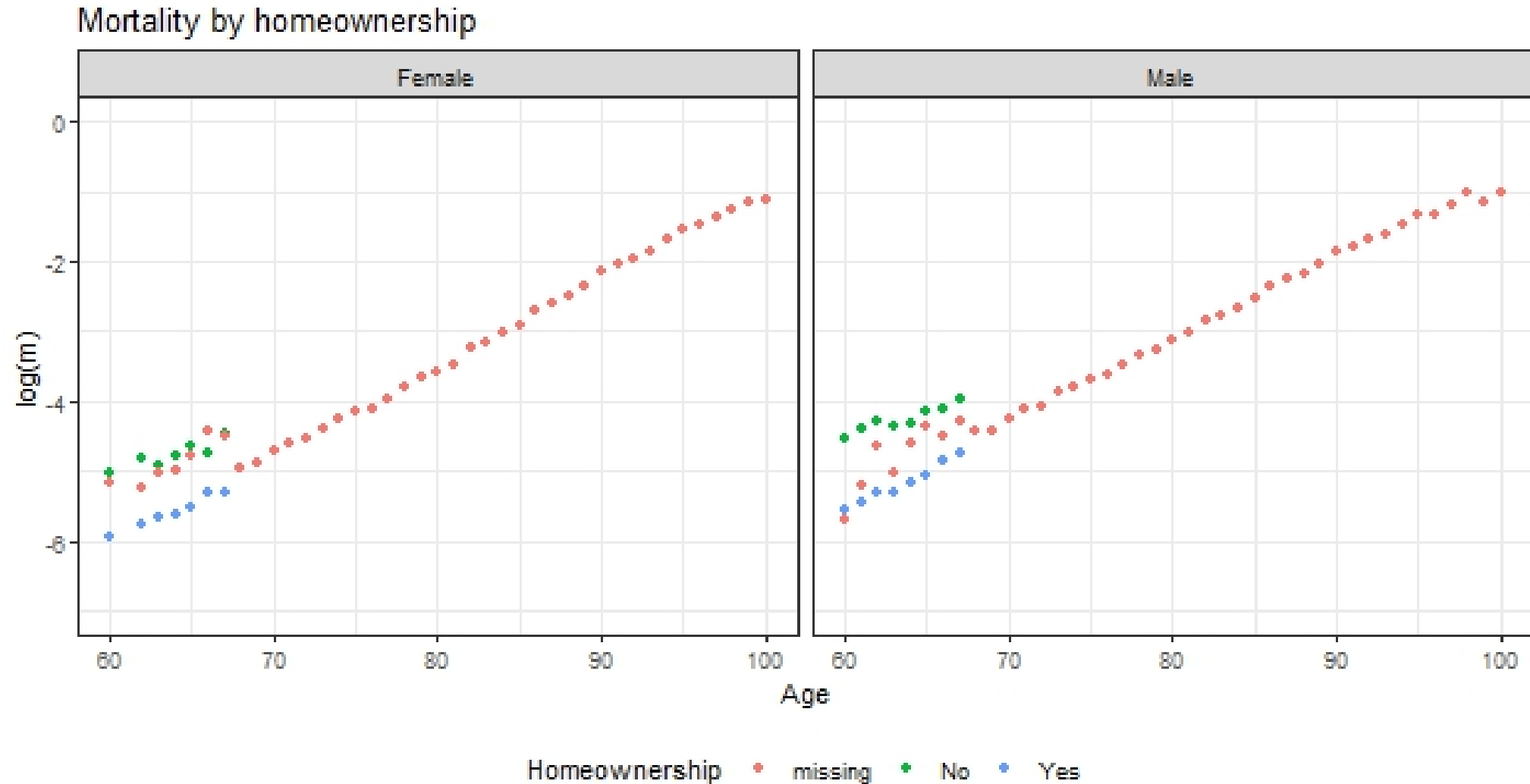


Income (Personal weekly income)

Mortality by personal income group



Homeownership

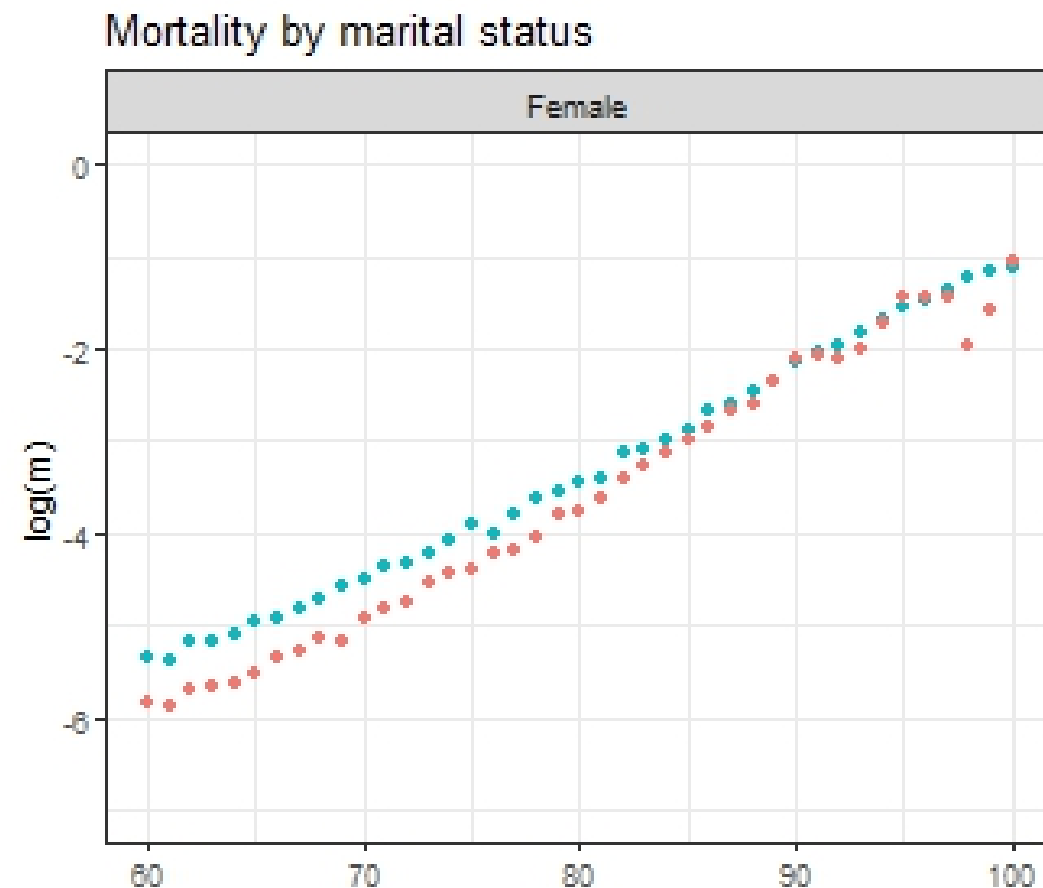
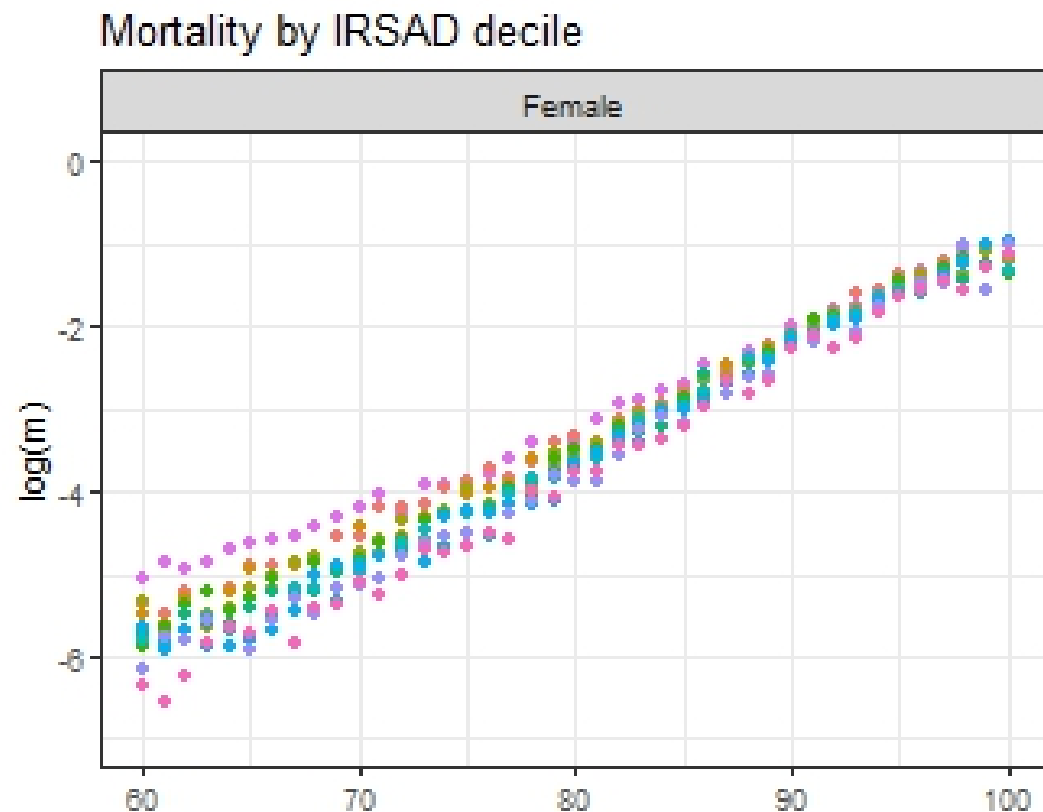


Our main goal

Develop a **parsimonious model** to produce life tables that **simultaneously** account for the **combined effects** of:

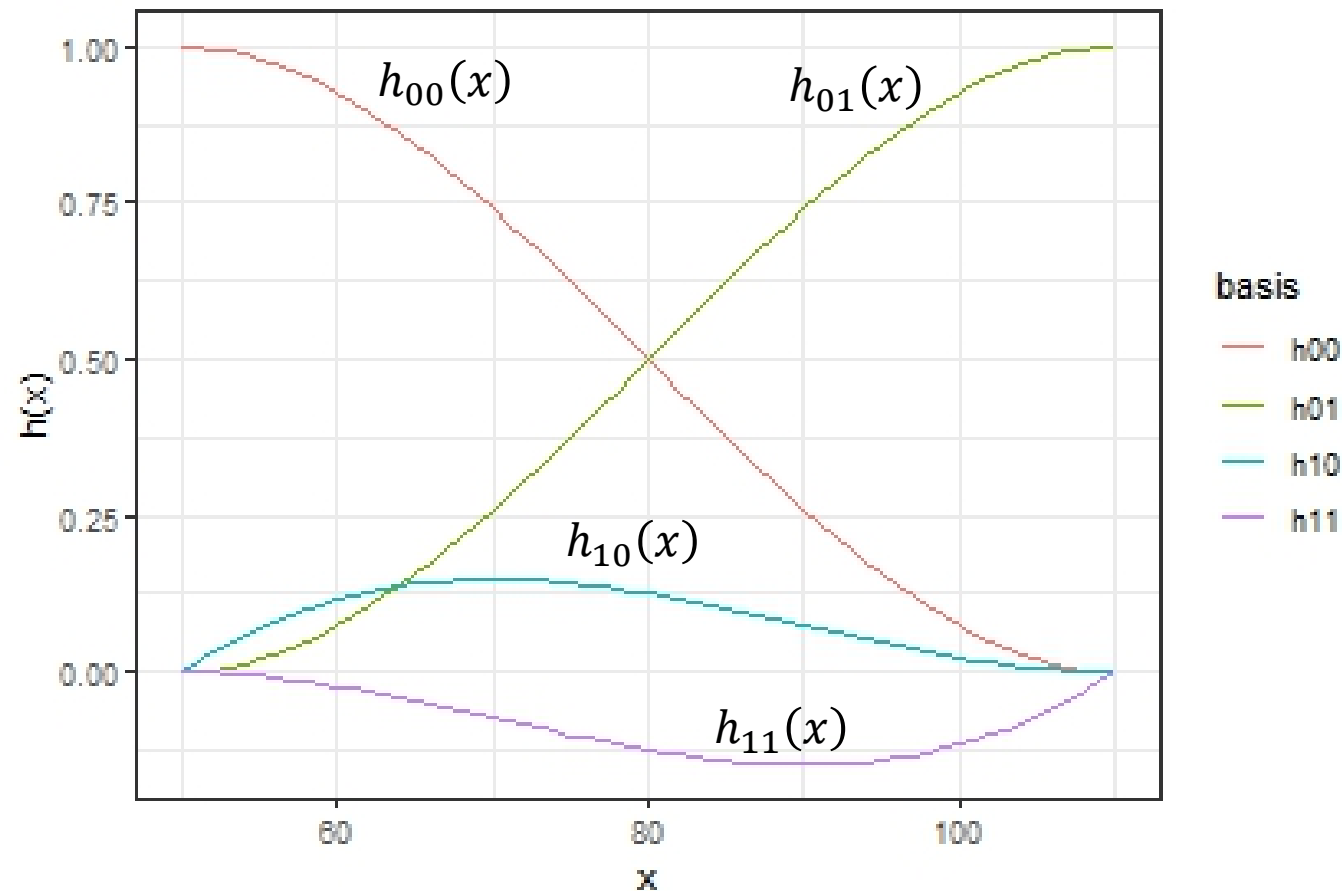
- Age
- Gender
- IRSAD
- Marital Status
- Income
- Homeownership

Diminishing differences in mortality with rising age



Hermite splines model (Richards, 2020)

$$\log \mu_x = ah_{00}(x) + \omega h_{01}(x) + m_0 h_{10}(x) + m_1 h_{11}(x)$$



Hermite splines model with covariates

$$\log \mu_{x,s,m,i,h} = ah_{00}(x) + \omega h_{01}(x) + m_0 h_{10}(x) + m_1 h_{11}(x) +$$

Covariate information

$\times h_{00}(x)$

Captures diminishing differences in mortality with rising age

Avoids mortality crossovers at old ages

Hermite splines model with covariates

$$\log \mu_{x,s,m,i,h} = ah_{00}(x) + \omega h_{01}(x) + m_0 h_{10}(x) + m_1 h_{11}(x) +$$

$$\left(\begin{array}{c} IRSAD (s) \\ + \\ Marital Status (m) \\ + \\ Income (i) \\ + \\ Homewonership (h) \end{array} \right) \times h_{00}(x)$$

Hermite splines model with covariates

$$\log \mu_{x,s,m,i,h} = ah_{00}(x) + \omega h_{01}(x) + m_0 h_{10}(x) + m_1 h_{11}(x) +$$

$$\left(\begin{array}{c} a_{IRSAD,D2}u_{s,m,i,h} + \cdots + a_{IRSAD,D10}u_{s,m,i,h} \\ + \\ a_{married}w_{s,m,i,h} \\ + \\ a_{income,500-999}z_{s,m,i,h} + a_{income,1000+}z_{s,m,i,h} + a_{income,missing}z_{s,m,i,h} \\ + \\ a_{homewoner,Yes}v_{s,m,i,h} + a_{homewoner,missing}v_{s,m,i,h} \end{array} \right) \times h_{00}(x)$$

19 parameters = 4 (age) + 9 (IRSAD) + 1 (Marital Status) + 3 (Income) + 2 (Homeownership)

240 Life Tables = 10 (IRSAD) x 2 (Marital Status) x 4 (Income) x 3 (Homeownership)

Coefficient	Male	Female
a	-3.96	-4.08
ω	-0.30	-0.24
m_0	-1.13	-4.79
m_1	2.23	3.12
$a_{IRSAD,D2}$	-0.14	-0.19
$a_{IRSAD,D3}$	-0.26	-0.37
$a_{IRSAD,D4}$	-0.37	-0.39
$a_{IRSAD,D5}$	-0.35	-0.45
$a_{IRSAD,D6}$	-0.51	-0.55
$a_{IRSAD,D7}$	-0.52	-0.61
$a_{IRSAD,D8}$	-0.57	-0.66
$a_{IRSAD,D9}$	-0.65	-0.79
$a_{IRSAD,D10}$	-0.75	-0.87
$a_{married}$	-0.53	-0.45
$a_{income,500-999}$	-0.53	-0.59
$a_{income,1000+}$	-0.96	-0.88
$a_{income,missing}$	1.16	1.37
$a_{homewoner,Yes}$	-0.51	-0.51
$a_{homewoner,missing}$	-0.34	-0.23

4 Age Parameters

9 IRSDA Parameters

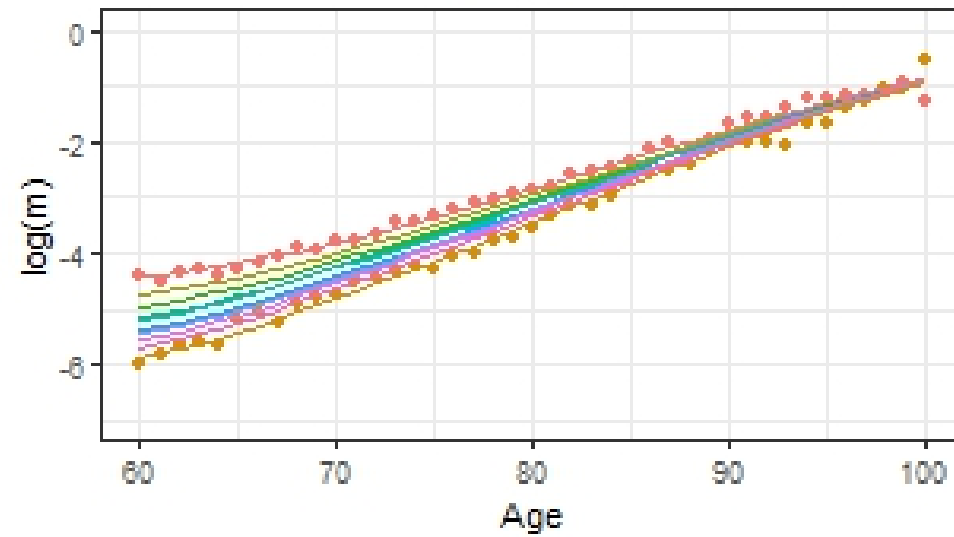
1 Marital Status Parameters

3 Income Parameters

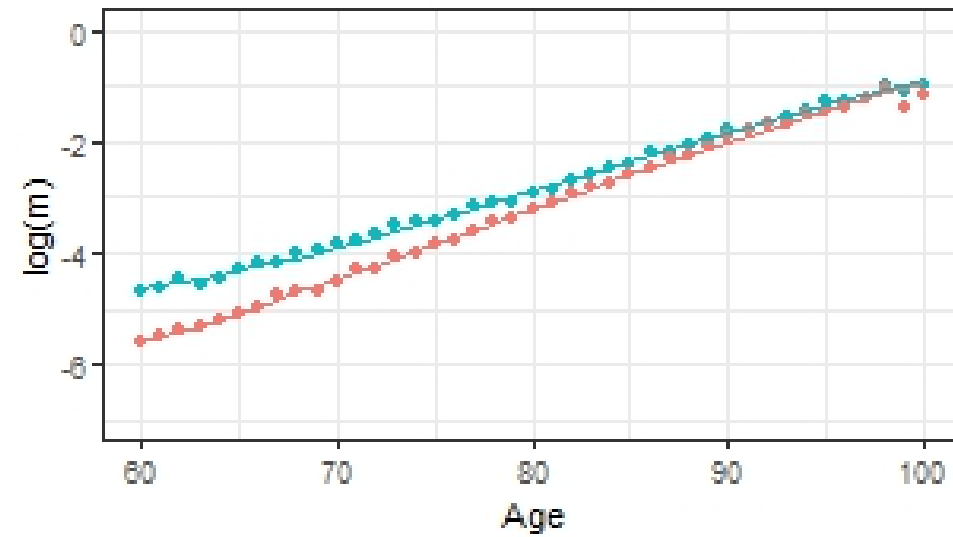
2 Homeownership Parameters

Fitted and Observed Rates for Males

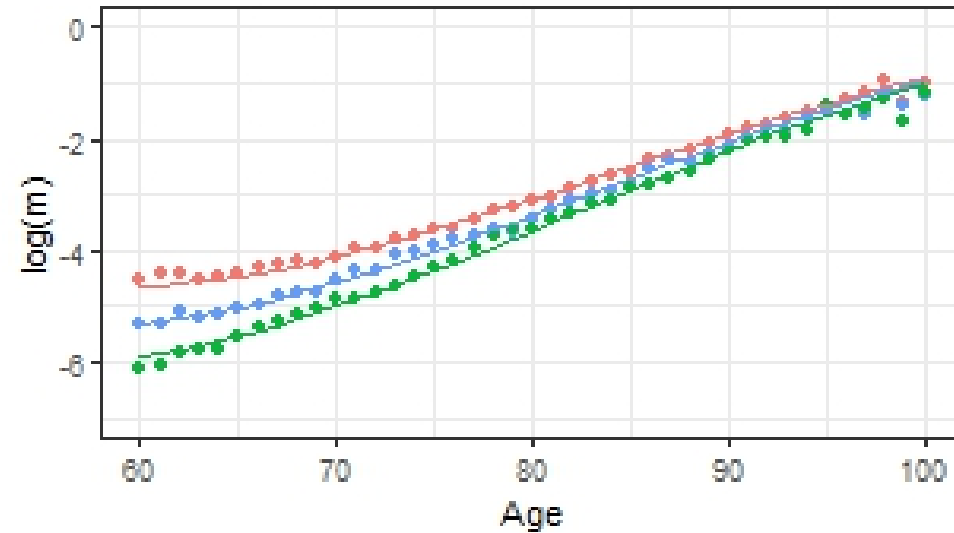
IRSAD



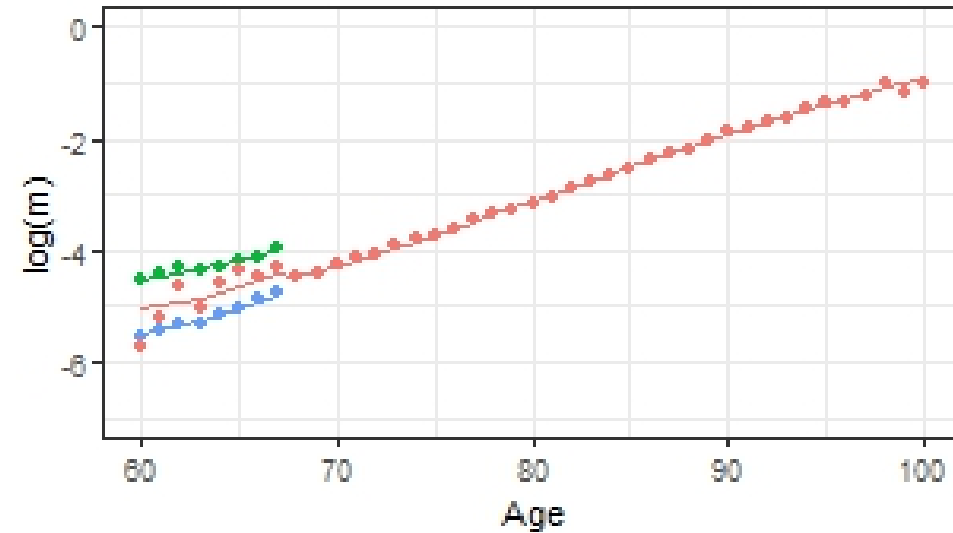
Marital status



Personal income group

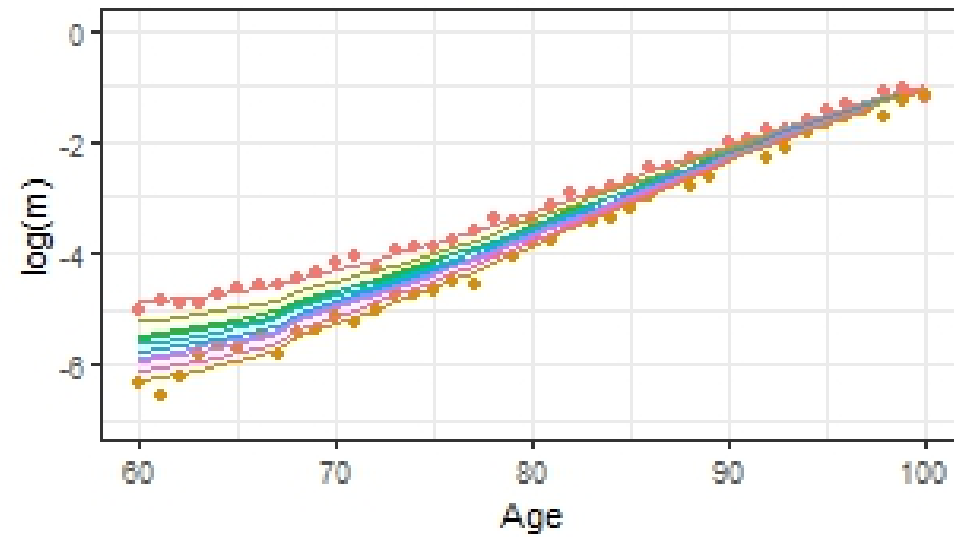


Homeownership

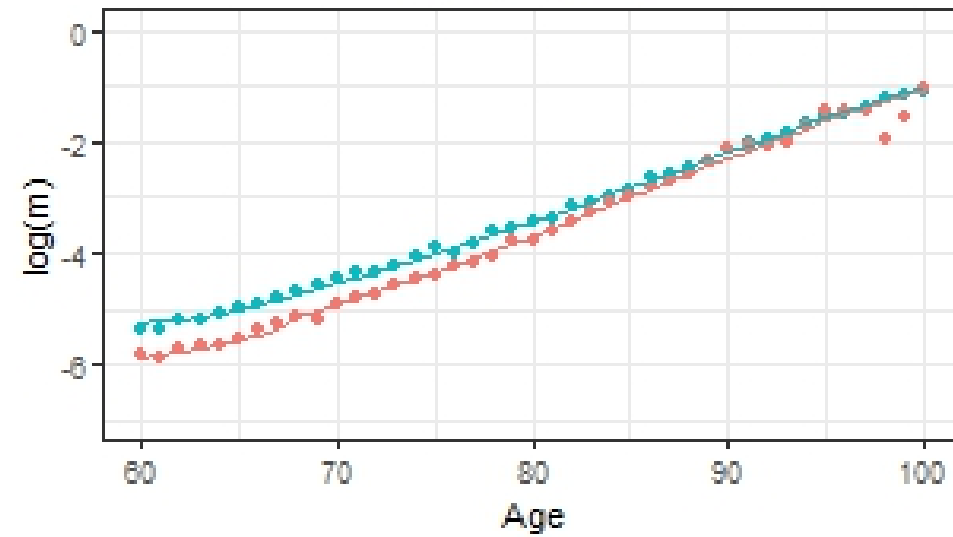


Fitted and Observed Rates for Females

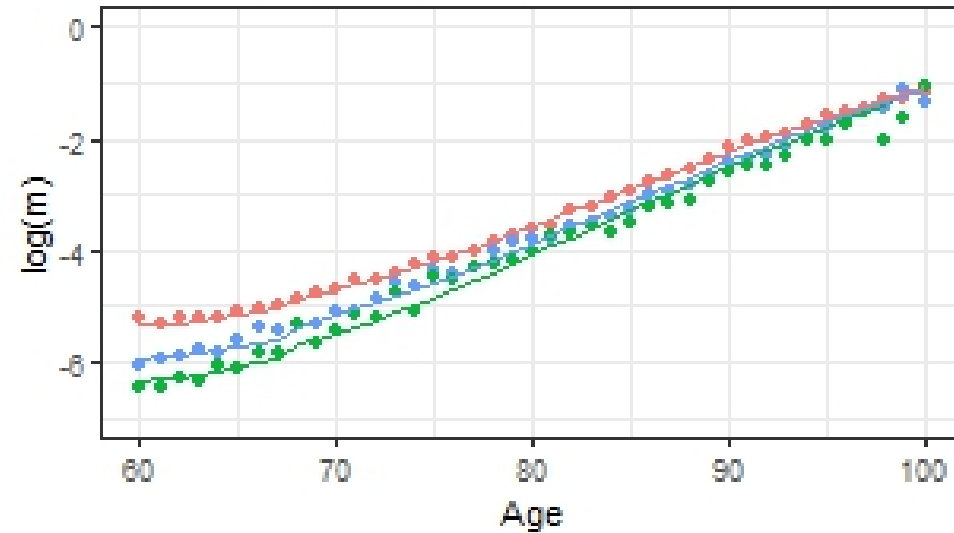
IRSAD



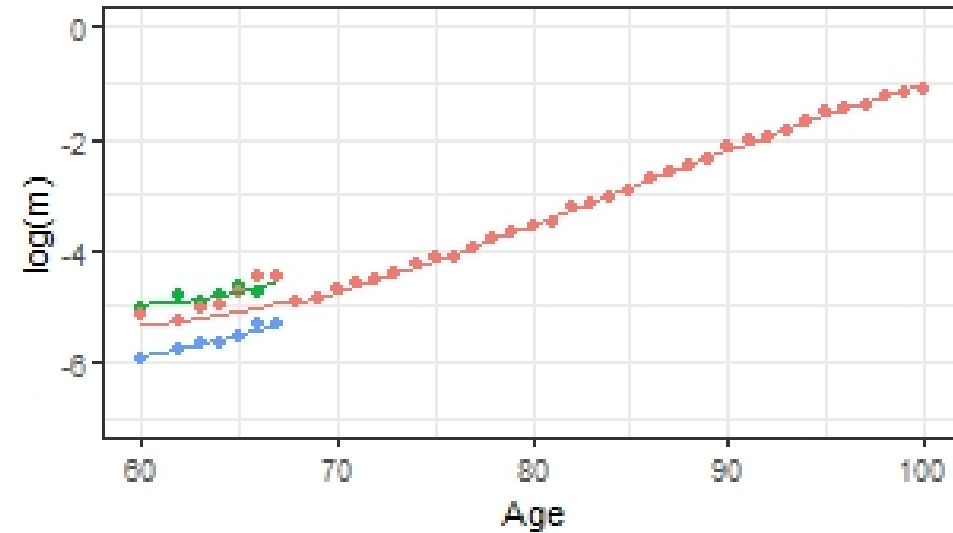
Marital status



Personal income group



Homeownership

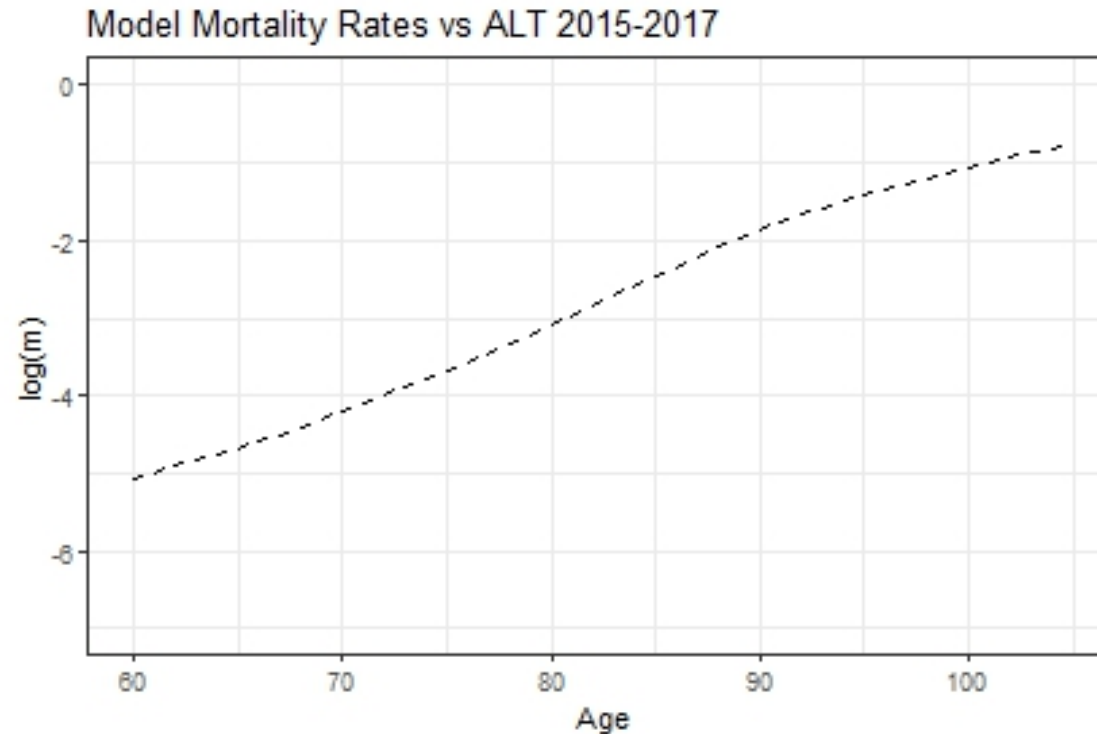


Model use

Coefficient	Male
a	-3.96
ω	-0.30
m_0	-1.13
m_1	2.23
$a_{IRSAD,D2}$	-0.14
$a_{IRSAD,D3}$	-0.26
$a_{IRSAD,D4}$	-0.37
$a_{IRSAD,D5}$	-0.35
$a_{IRSAD,D6}$	-0.51
$a_{IRSAD,D7}$	-0.52
$a_{IRSAD,D8}$	-0.57
$a_{IRSAD,D9}$	-0.65
$a_{IRSAD,D10}$	-0.75
$a_{married}$	-0.53
$a_{income,500-999}$	-0.53
$a_{income,1000+}$	-0.96
$a_{income,missing}$	1.16
$a_{homewoner,Yes}$	-0.51
$a_{homewoner,missing}$	-0.34

Homeownership	Marital Status	Weekly Income	IRSAD

$$\log \mu_x = ah_{00}(x) + \omega h_{01}(x) + m_0 h_{10}(x) + m_1 h_{11}(x)$$



**Period Life Expectancy
in 2016 at 60**

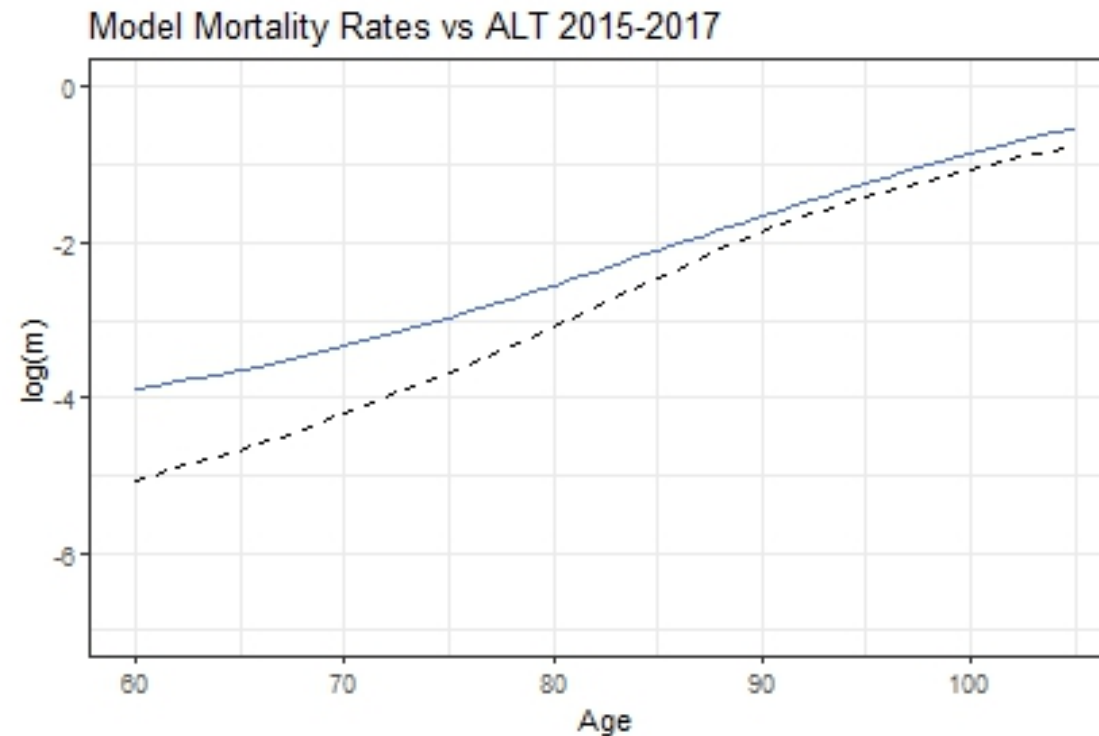
ALT 2015-2017: 24.05

Model use

Coefficient	Male
a	-3.96
ω	-0.30
m_0	-1.13
m_1	2.23
$a_{IRSAD,D2}$	-0.14
$a_{IRSAD,D3}$	-0.26
$a_{IRSAD,D4}$	-0.37
$a_{IRSAD,D5}$	-0.35
$a_{IRSAD,D6}$	-0.51
$a_{IRSAD,D7}$	-0.52
$a_{IRSAD,D8}$	-0.57
$a_{IRSAD,D9}$	-0.65
$a_{IRSAD,D10}$	-0.75
$a_{married}$	-0.53
$a_{income,500-999}$	-0.53
$a_{income,1000+}$	-0.96
$a_{income,missing}$	1.16
$a_{homewoner,Yes}$	-0.51
$a_{homewoner,missing}$	-0.34

Homeownership	Marital Status	Weekly Income	IRSAD
No	Single	\$499 or less	D1 (most disadvantaged)

$$\log \mu_x = -3.96h_{00}(x) - 0.3h_{01}(x) - 1.13h_{10}(x) + 2.23h_{11}(x)$$



**Period Life Expectancy
in 2016 at 60**

ALT 2015-2017: 24.05

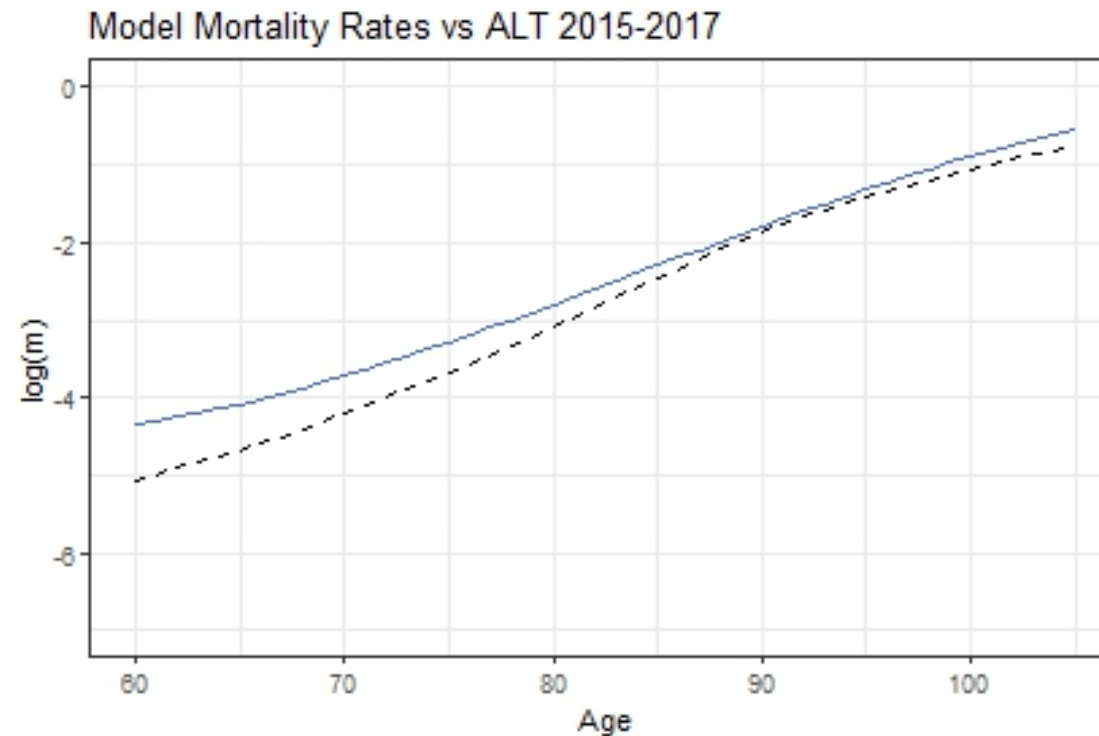
Model: 18.67

Model use

Coefficient	Male
a	-3.96
ω	-0.30
m_0	-1.13
m_1	2.23
$a_{IRSAD,D2}$	-0.14
$a_{IRSAD,D3}$	-0.26
$a_{IRSAD,D4}$	-0.37
$a_{IRSAD,D5}$	-0.35
$a_{IRSAD,D6}$	-0.51
$a_{IRSAD,D7}$	-0.52
$a_{IRSAD,D8}$	-0.57
$a_{IRSAD,D9}$	-0.65
$a_{IRSAD,D10}$	-0.75
$a_{married}$	-0.53
$a_{income,500-999}$	-0.53
$a_{income,1000+}$	-0.96
$a_{income,missing}$	1.16
$a_{homewoner,Yes}$	-0.51
$a_{homewoner,missing}$	-0.34

Homeownership	Marital Status	Weekly Income	IRSAD
Yes	Single	\$499 or less	D1 (most disadvantaged)

$$\log \mu_x = -4.47h_{00}(x) - 0.3h_{01}(x) - 1.13h_{10}(x) + 2.23h_{11}(x)$$



**Period Life Expectancy
in 2016 at 60**

ALT 2015-2017: 24.05

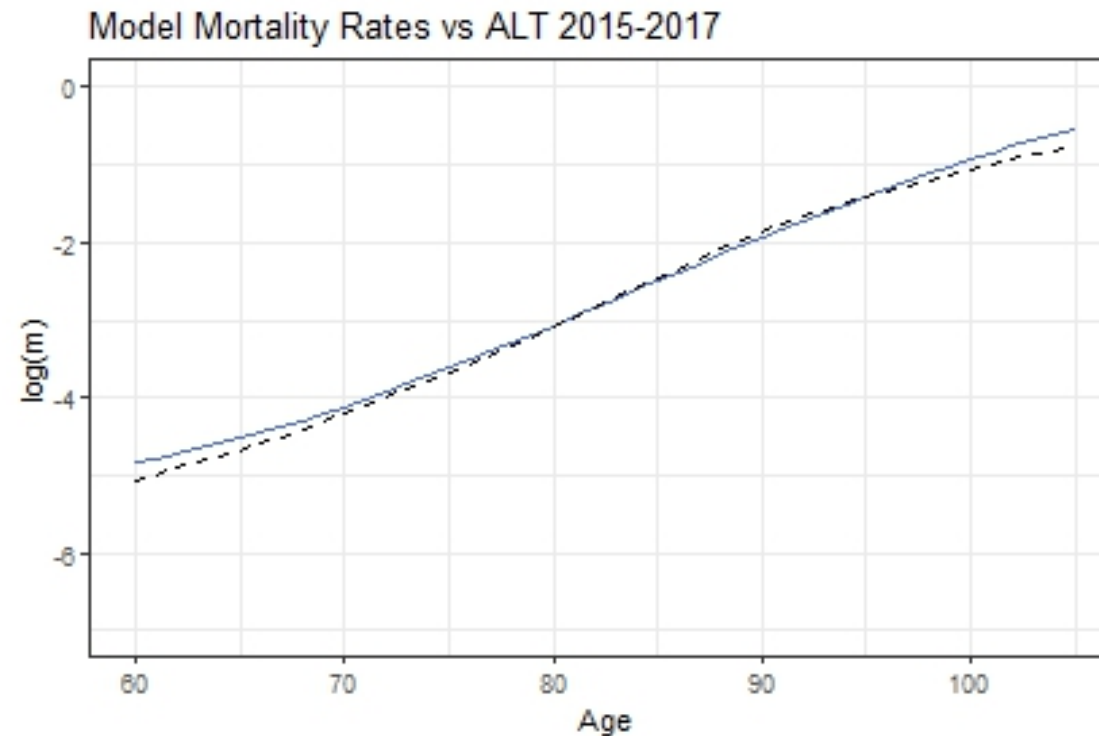
Model: 21.45

Model use

Coefficient	Male
a	-3.96
ω	-0.30
m_0	-1.13
m_1	2.23
$a_{IRSAD,D2}$	-0.14
$a_{IRSAD,D3}$	-0.26
$a_{IRSAD,D4}$	-0.37
$a_{IRSAD,D5}$	-0.35
$a_{IRSAD,D6}$	-0.51
$a_{IRSAD,D7}$	-0.52
$a_{IRSAD,D8}$	-0.57
$a_{IRSAD,D9}$	-0.65
$a_{IRSAD,D10}$	-0.75
$a_{married}$	-0.53
$a_{income,500-999}$	-0.53
$a_{income,1000+}$	-0.96
$a_{income,missing}$	1.16
$a_{homewoner,Yes}$	-0.51
$a_{homewoner,missing}$	-0.34

Homeownership	Marital Status	Weekly Income	IRSAD
Yes	Married	\$499 or less	D1 (most disadvantaged)

$$\log \mu_x = -5.00h_{00}(x) - 0.3h_{01}(x) - 1.13h_{10}(x) + 2.23h_{11}(x)$$



**Period Life Expectancy
in 2016 at 60**

ALT 2015-2017: 24.05

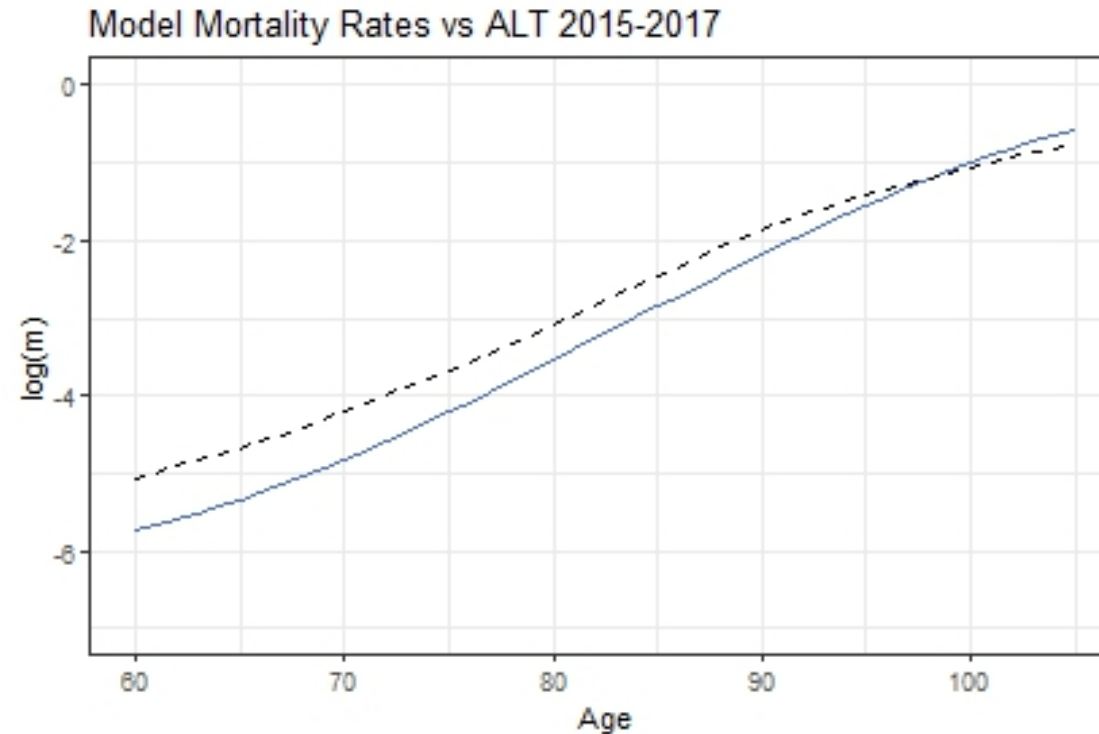
Model: 24.06

Model use

Coefficient	Male
a	-3.96
ω	-0.30
m_0	-1.13
m_1	2.23
$a_{IRSAD,D2}$	-0.14
$a_{IRSAD,D3}$	-0.26
$a_{IRSAD,D4}$	-0.37
$a_{IRSAD,D5}$	-0.35
$a_{IRSAD,D6}$	-0.51
$a_{IRSAD,D7}$	-0.52
$a_{IRSAD,D8}$	-0.57
$a_{IRSAD,D9}$	-0.65
$a_{IRSAD,D10}$	-0.75
$a_{married}$	-0.53
$a_{income,500-999}$	-0.53
$a_{income,1000+}$	-0.96
$a_{income,missing}$	1.16
$a_{homewoner,Yes}$	-0.51
$a_{homewoner,missing}$	-0.34

Homeownership	Marital Status	Weekly Income	IRSAD
Yes	Married	\$1000 or more	D1 (most disadvantaged)

$$\log \mu_x = -5.96h_{00}(x) - 0.3h_{01}(x) - 1.13h_{10}(x) + 2.23h_{11}(x)$$



**Period Life Expectancy
in 2016 at 60**

ALT 2015-2017: 24.05

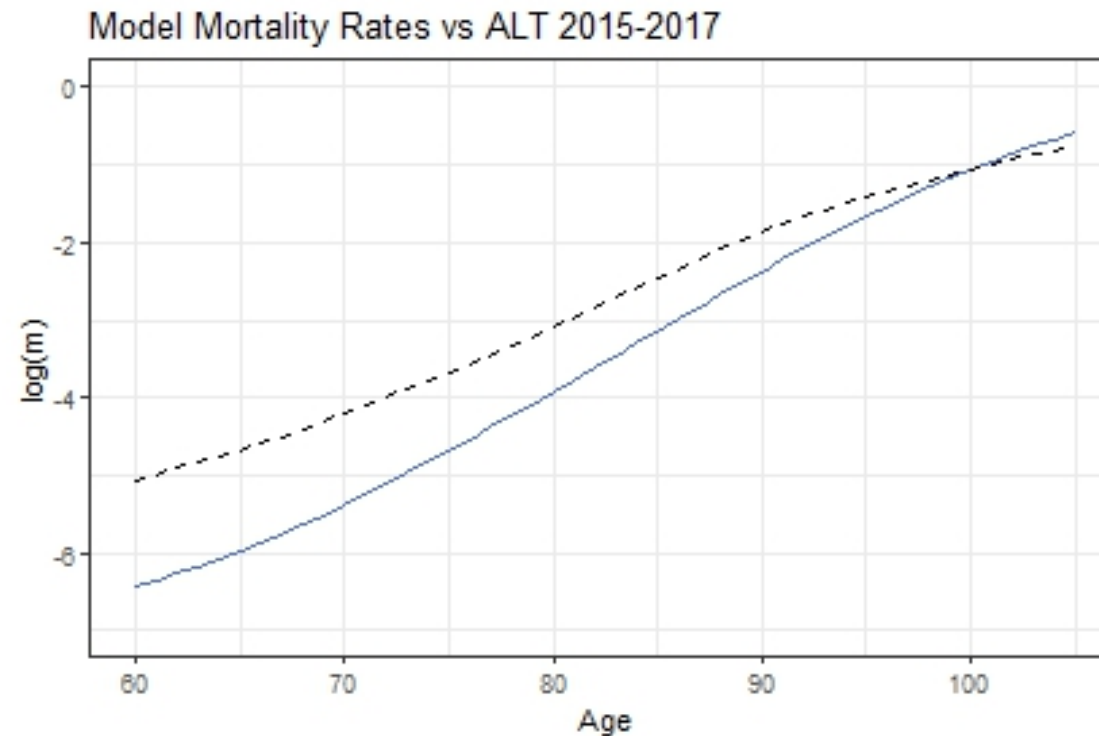
Model: 27.89

Model use

Coefficient	Male
a	-3.96
ω	-0.30
m_0	-1.13
m_1	2.23
$a_{IRSAD,D2}$	-0.14
$a_{IRSAD,D3}$	-0.26
$a_{IRSAD,D4}$	-0.37
$a_{IRSAD,D5}$	-0.35
$a_{IRSAD,D6}$	-0.51
$a_{IRSAD,D7}$	-0.52
$a_{IRSAD,D8}$	-0.57
$a_{IRSAD,D9}$	-0.65
$a_{IRSAD,D10}$	-0.75
$a_{married}$	-0.53
$a_{income,500-999}$	-0.53
$a_{income,1000+}$	-0.96
$a_{income,missing}$	1.16
$a_{homewoner,Yes}$	-0.51
$a_{homewoner,missing}$	-0.34

Homeownership	Marital Status	Weekly Income	IRSAD
Yes	Married	\$1000 or more	D10 (most Advantaged)

$$\log \mu_x = -6.71h_{00}(x) - 0.3h_{01}(x) - 1.13h_{10}(x) + 2.23h_{11}(x)$$

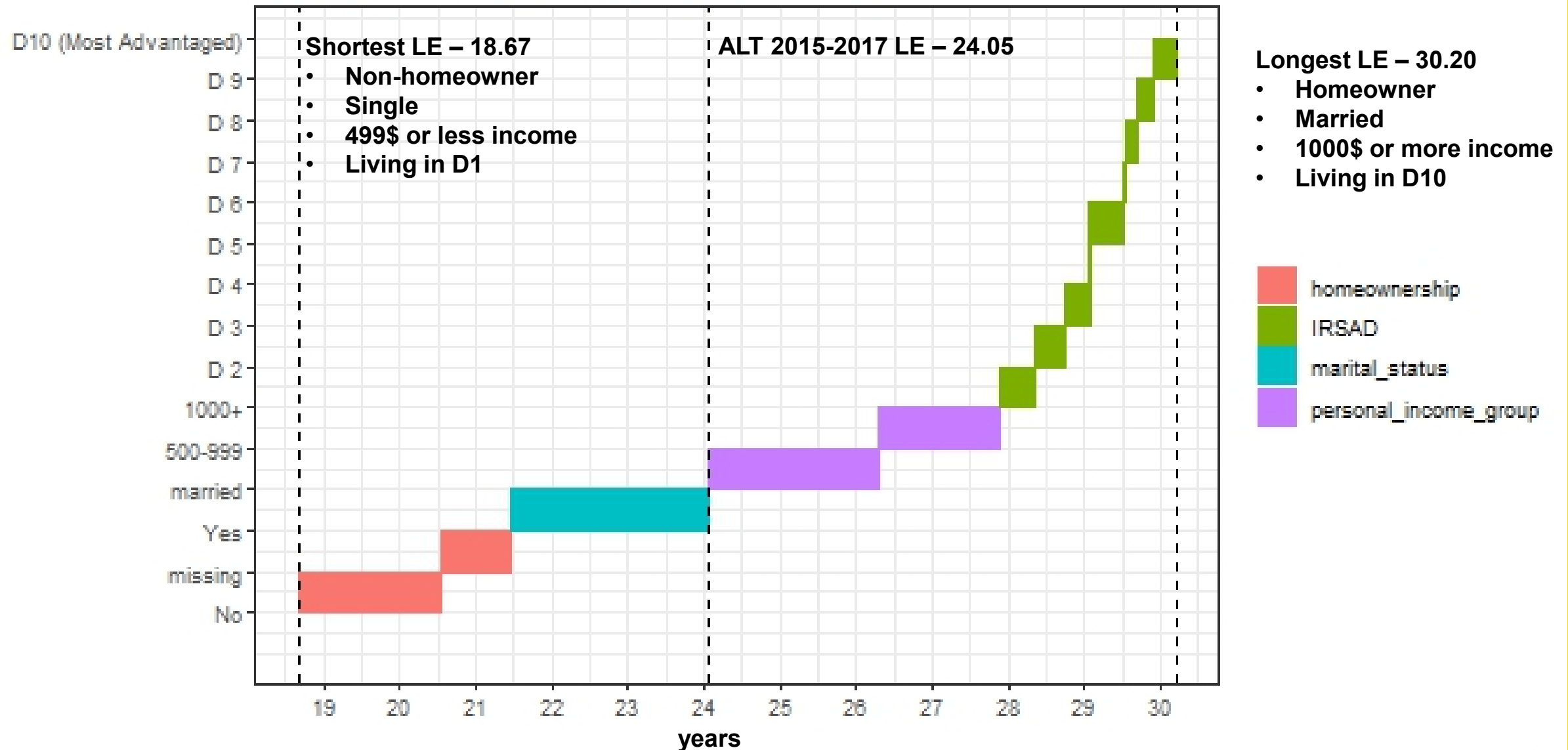


**Period Life Expectancy
in 2016 at 60**

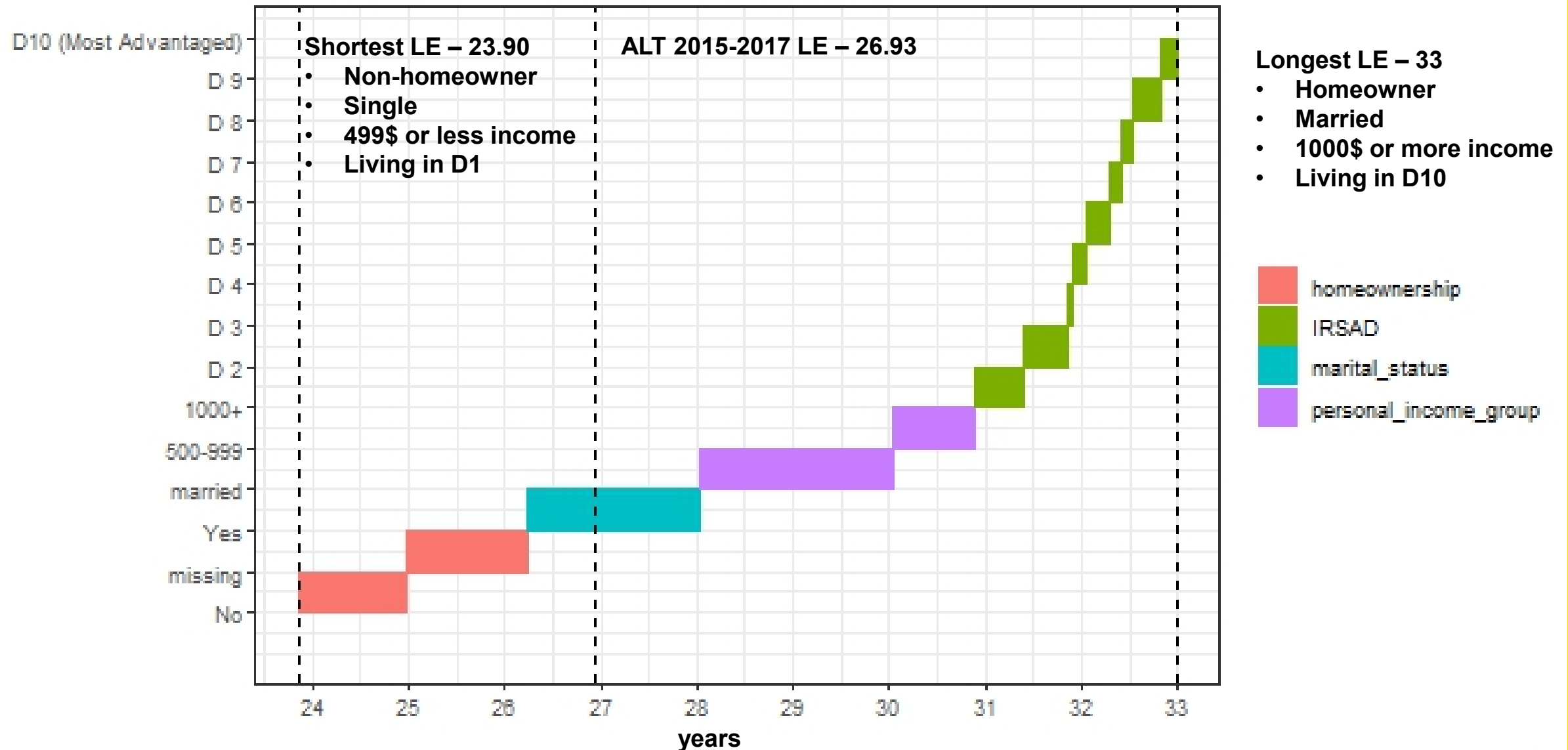
ALT 2015-2017: 24.05

Model: 30.20

Period Life Expectancy at 60 in 2016 for males



Period Life Expectancy at 60 in 2016 for females



Annuity Income and Retirement Products Implications

Annuity Income Variation:

For a \$100,000 annuity at age 65 (3% interest, no indexation or death benefit):

- Lowest life expectancy: \$8,521 (males), \$6,896 (females)
- Highest life expectancy: \$5,785 (males), \$5,387 (females)
- A disadvantaged female could receive 28% higher annuity if differences are reflected in pricing.

Equity in Retirement Products:

- Uniform pricing of longevity products may unintentionally disadvantage lower socio-economic groups.
- Annuity design (e.g., death benefit, indexation, guarantee period) can influence the size of these income differentials.

Resources: Paper

- The paper is available on SSRN.

Towards Fairer Retirement Outcomes: Socio-Economic Mortality Differentials in Australia

38 Pages • Posted: 14 May 2025 • Last revised: 1 Jun 2025

[Fei Huang](#)

UNSW Australia Business School, School of Risk & Actuarial Studies

[Francis Hui](#)

The Australian National University

[Andrés Villegas](#)

University of New South Wales (UNSW) - ARC Centre of Excellence in Population Ageing Research (CEPAR)

Date Written: May 14, 2025

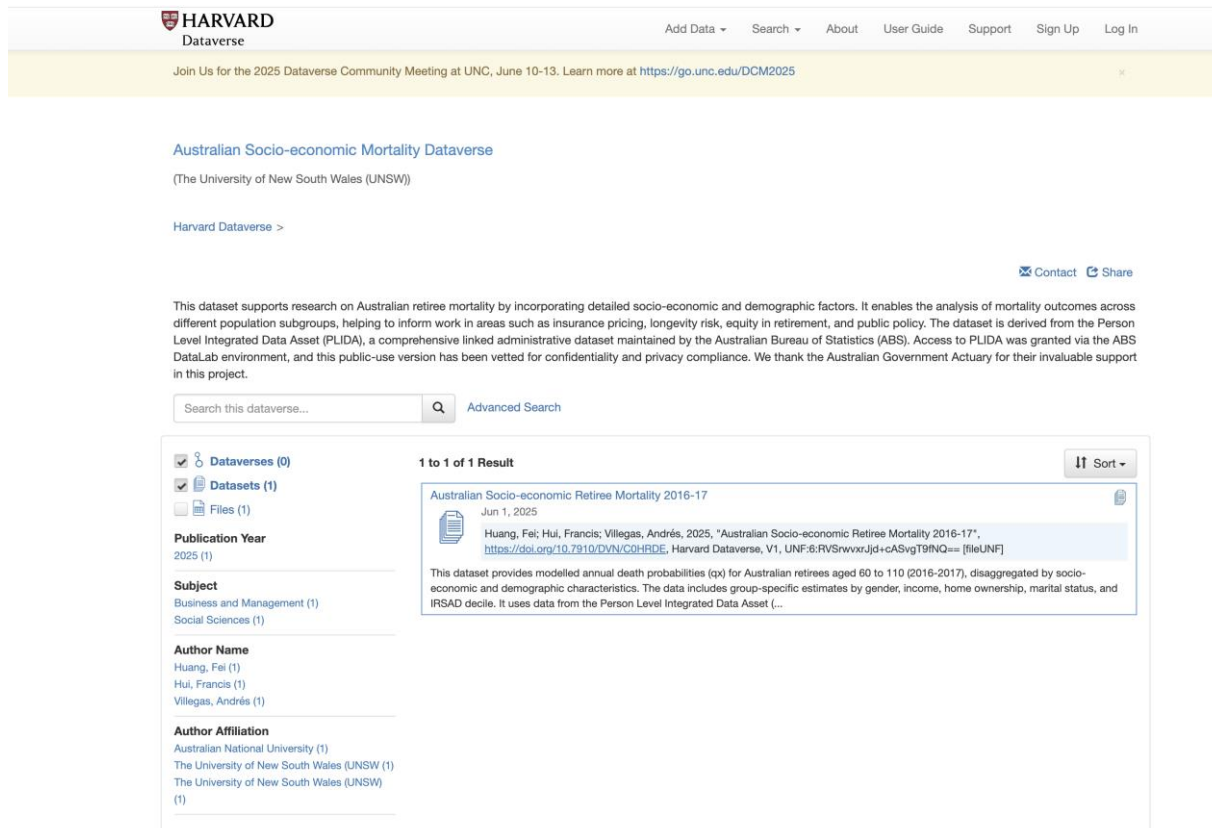
Abstract

The Australian superannuation system faces the challenge of delivering equitable and sustainable income in retirement, yet the development of effective longevity products remains limited. A key barrier is the lack of detailed understanding of mortality differentials across socioeconomic groups, which complicates the pricing and design of retirement income products. While it is known that mortality varies by factors such as income and education, comprehensive evidence for Australia has been scarce, particularly for the post-retirement population. This paper addresses this gap by analysing individual-level linked data from the Australian Bureau of Statistics' Personal Level Integrated Data Asset (PLIDA), covering the entire Australian population aged 60-100 over 2016-2017. We examine mortality differentials across socioeconomic indicators including area-level socioeconomic advantage and disadvantage, income, marital status, and home ownership using flexible Hermite-spline Poisson regression models. Our results reveal substantial disparities in mortality and life expectancy across socioeconomic groups, with differences narrowing at older ages. For example, at age 60, the gap in period life expectancy between the most and least advantaged males is 11.5 years, and 9.1 years for females. Longevity differences translate into substantial variation in annuity income. The large and persistent differences by socioeconomic characteristics suggest that uniform approaches to longevity product pricing may unintentionally disadvantage certain groups. These findings highlight the importance of accounting for heterogeneity in longevity when designing retirement income products and inform policies aimed at fairer outcomes in the retirement phase.



Resources: Open Dataset “Australian Socio-Economic Mortality Dataverse”

- The dataset on the modelled annual probability of death (q_x) for ages 60–100 is openly available on the Harvard Dataverse: Australian Socio-economic Mortality Dataverse.



The screenshot displays the Harvard Dataverse interface. At the top, the Harvard Dataverse logo is visible alongside navigation links: Add Data, Search, About, User Guide, Support, Sign Up, and Log In. A banner below the header invites users to join the 2025 Dataverse Community Meeting at UNC, June 10-13, with a link to <https://go.unc.edu/DCM2025>.

The main content area features the title "Australian Socio-economic Mortality Dataverse" and its origin, "(The University of New South Wales (UNSW))". Below this, a link to "Harvard Dataverse >" is provided. A "Contact" and "Share" button are also present.

A descriptive paragraph states: "This dataset supports research on Australian retiree mortality by incorporating detailed socio-economic and demographic factors. It enables the analysis of mortality outcomes across different population subgroups, helping to inform work in areas such as insurance pricing, longevity risk, equity in retirement, and public policy. The dataset is derived from the Person Level Integrated Data Asset (PLIDA), a comprehensive linked administrative dataset maintained by the Australian Bureau of Statistics (ABS). Access to PLIDA was granted via the ABS DataLab environment, and this public-use version has been vetted for confidentiality and privacy compliance. We thank the Australian Government Actuary for their invaluable support in this project."

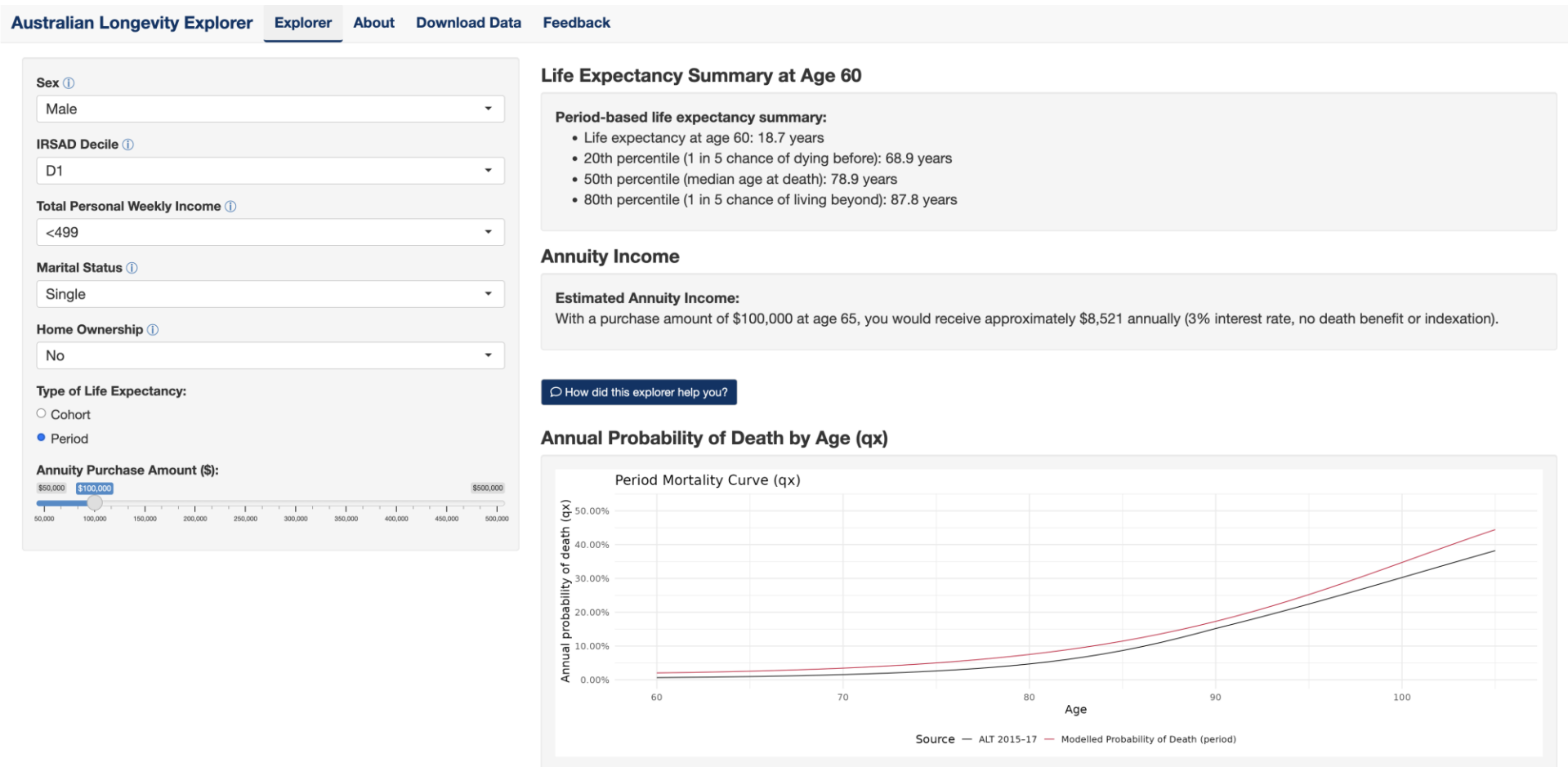
A search bar with the placeholder "Search this dataverse..." and an "Advanced Search" link are located below the description.

The search results section, titled "1 to 1 of 1 Result", shows a single entry: "Australian Socio-economic Retiree Mortality 2016-17". The entry includes a document icon, the date "Jun 1, 2025", the authors "Huang, Fei; Hui, Francis; Villegas, Andrés, 2025", the title "Australian Socio-economic Retiree Mortality 2016-17", and a DOI link: <https://doi.org/10.7910/DVN/COHRDE>. A brief description follows: "This dataset provides modelled annual death probabilities (q_x) for Australian retirees aged 60 to 110 (2016-2017), disaggregated by socio-economic and demographic characteristics. The data includes group-specific estimates by gender, income, home ownership, marital status, and IRSAD decile. It uses data from the Person Level Integrated Data Asset (...)"

On the left side of the results, a sidebar lists filters: "Dataverses (0)", "Datasets (1)", and "Files (1)". Below these, filters for "Publication Year" (2025 (1)), "Subject" (Business and Management (1), Social Sciences (1)), "Author Name" (Huang, Fei (1), Hui, Francis (1), Villegas, Andrés (1)), and "Author Affiliation" (Australian National University (1), The University of New South Wales (UNSW) (1), The University of New South Wales (UNSW) (1)) are shown.



Resources: An Interactive App - “Australian Longevity Explorer”



<https://auslongevity.shinyapps.io/explorer/>

Conclusion

- Important differences in mortality and life expectancy by socio-economic variables
- Parsimonious model that allows quantification of mortality differentials of 240 different socioeconomic profiles
- Differences in period life expectancy at age 60 of 12 years for males and 9 years for females accounting for IRSAD, Marital Status, Income and Homeownership
- These differences should be reflected when designing retirement income products

Future work

- Update of model using 2021 Census data
- Incorporating Health Data (MBS and PBS) to understand the mortality differentials

Thank you!

a.villegas@unsw.edu.au



Scan to get our working paper

References

Australian Bureau of Statistics. “Person Level Integrated Data Asset (PLIDA), ABS DataLab.” *Findings Based on Use of PLIDA Data*, Accessed 2023.

Richards, Stephen J. 2020. “A Hermite-Spline Model of Post-Retirement Mortality.” *Scandinavian Actuarial Journal* 2020 (2): 110–27.

Marital Status

- Based on **Registered marital status (MSTP)** on 2016 Census
- **MSTP codes**
 - 1 Never married
 - 2. Widowed
 - 3 Divorced
 - 4 Separated
 - 5 Married
 - @ Not applicable
- **Single:** 1,2,3,4
- **Married:** 5

What is Person 1's current marital status?

'Married' refers to registered marriages.

▶ [More information](#)

- ☐ Never married
- ☐ Widowed
- ☐ Divorced
- ☐ Separated but not divorced
- ☐ Married

Income

- Based on **Total personal income (weekly) (INCP)** on 2016 Census
- **Personal income group**
 - <499\$
 - 500\$-999\$
 - 1000\$+
 - Missing

What is the *total* of all income Person 1 *usually* receives?

Do not deduct: tax, superannuation contributions, amounts salary sacrificed, or any other automatic deductions.

Include:

- Wages and salaries
 - Regular overtime
 - Commissions and bonuses
- Government pensions, benefits and allowances
- Profit or loss from:
 - Unincorporated business/farm (e.g. sole traders, partnerships)
 - Rental properties
- Other income from:
 - Superannuation
 - Child support
 - Dividends from shares
 - Interest
 - Workers' compensation
 - Any other income sources

Homeownership

- Two sources of data
 - **Tenure and landlord type (TENLLD)** on 2016 Census
 - **Social Security Related Information (SSRI)**
- **Homeownership**
 - Homeowner
 - Non-homeowner
 - Missing
- Homeownership information **only usable before age 68**
 - Census refers to location on census night and includes people in hospital and in age-cared facilities

Categories:

- 1 Owned outright
 - 2 Owned with a mortgage
 - 3 Rented: Real estate agent
 - 4 Rented: State or territory housing authority
 - 5 Rented: Person not in same household
 - 6 Rented: Housing co-operative, community or church group
 - 7 Rented: Other landlord type
 - 8 Rented: Landlord type not stated
 - 9 Other tenure type
 - & Tenure type not stated
 - @ Tenure type not applicable
-

Number of categories: 11

Not applicable (@) category comprises:

- Unoccupied private dwellings
- Non-private dwellings
- Migratory, off-shore and shipping SA1s